Academic Year 2025-2026	**FACULTY OF SCIENCES OF GAFSA** **Computer Science Department**	**Instructor** Dr. **N**adia **S**MAIRI
Section : Master	**M**achine **L**earning	Project

Machine learning Project

The field of cybersecurity has witnessed a growing emphasis on the integration of machine learning and deep learning techniques, as researchers seek to address the challenges posed by the ever-increasing volume and complexity of data. Cybersecurity data science has garnered substantial attention, with researchers exploring the potential of intelligent models to enhance the identification and protection against cyber threats. In particular, Intrusion Detection Systems (IDS) have become a central application area where machine learning is used to automatically monitor network traffic and detect malicious or abnormal activities. These technologies offer the ability to adapt to the dynamic landscape of cyber threats, which often involve intricate and interconnected patterns that are difficult to capture using traditional rule-based approaches. Various machine learning techniques, including classification, regression, security data clustering, and rule-based modeling, have demonstrated their effectiveness in developing advanced IDS and cybersecurity solutions for diverse applications.

The aim of this project is to respond to this query : How can modern machine learning and deep learning techniques improve the performance of IDS using benchmark datasets such as NSL-KDD? Thus, our primary object is to develop and evaluate a deep learning-based intrusion detection system using the NSL-KDD dataset.

Our project has three phases:

I. Phase 1:

- Introduction to cybersecurity and IDS fundamentals.
- Presentation of IDS challenges.
- State of the art of IDS datasets.
- Evaluation metrics used in IDS.

⇨ **Due date Tuesday 3rd March 2026.**

II. Phase 2:

- State of the art of existing methods (with a focus on deep learning methods from the last 5 years).
- Students must choose a recent survey and extract its taxonomy.
- Preparing a final comparative table:

Paper	Year	Dataset	Model	Results	Contribution	Limitations

- Presenting a research gap section.

⇨ **Due date Tuesday 31 March 2026.**

III. Phase 3:

- Implementation of a chosen deep learning method using NSL-KDD dataset.

⇨ **Due date Tuesday 28 April 2026.**